

CENTRAL FERTIGATION, STOCK-SOLUTION PREPARATION, EC/pH DOSING AND FERTILIZER MIXING

Activity-Specific Construction and Operating Method Statement

Document field	Controlled value
Document code	HF-GH01-MST-FER-001
Revision	00 - Draft for coordination
Project	One-Feddán NFT Lettuce Greenhouse Project
Location	Alexandria, Egypt
Served circuits	East 10 m3; west 10 m3; nursery 3 m3
Approved supplier concept	One complete Geneina fertigation/control package
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ONE TANK AT A TIME: Manual manifold selection and the controller recipe shall identify the same tank. A and B concentrates are never injected simultaneously or mixed in concentrated form.

1. Purpose and System Boundary

The system serves east production, west production and nursery NFT tanks through one central side-stream fertigation skid. Germination remains independent.

Circuit	Tank	Capacity	Interface	Program
EP	East production	10 m3	1.5 in tank nozzles reduced to 1 in loop	Production East recipe
WP	West production	10 m3	1.5 in tank nozzles reduced to 1 in loop	Production West recipe
NUR	Nursery NFT	3 m3	1 in suction and return loop	Nursery recipe
Excluded	Germination	1 m3	Independent local preparation	Not connected to central fertigation

Package and anti-duplication rules

Boundary	Adopted rule
Included inside Geneina package	Controller/panel, three dosing pumps, calibration columns, injection quills, internal check/back-pressure valves, side-stream pump, internal filter, gauges, rotameter, flow cell, EC/pH/temperature sensors, flush/drain valves and skid frame.
External supporting items	A/B/acid stock tanks, two mixers, suction assemblies, low-level switches, selection manifold, external pipework, preparation table, scale, sink, canopy, handheld meter and calibration kit.
Not duplicated	No separate pricing of internal Geneina components unless the signed supplier quotation explicitly excludes an item.
Explicitly excluded	Static mixer, separate stock-tank sight gauges, stock preparation water meter and permanent constructed chemical bund.

No direct plant-line injection: dosing enters the selected tank side-stream loop and returns to the same tank; injection into the buried 3 in plant feed is prohibited.

2. Equipment and Minimum Requirements

Code	Equipment	Qty	Minimum requirement
FER-001	Geneina fertigation/control package	1 set	Three dosing channels, internal 3-5 m3/h side-stream recirculation, EC/pH/temperature flow cell, controller, panel and skid.
FER-002	Stock solution tank A	1	500 L opaque chemical-resistant HDPE/PE, sealed lid and calibrated volume marks.
FER-003	Stock solution tank B	1	500 L opaque chemical-resistant HDPE/PE, sealed lid and calibrated volume marks.
FER-004	Acid / pH-down tank	1	100 L opaque acid-compatible HDPE/PE, sealed lid; no mixer.
FER-005	Stock-tank mixers	2	0.37 kW minimum, IP55 motor, SS316L or polymer-coated shaft/impeller; A and B only.
FER-006	Chemical suction assemblies	3 sets	Foot valve, cleanable strainer, isolation and chemical-compatible suction tube per A/B/acid tank.
FER-007	Stock low-level switches	3	One dry-contact low-level switch per stock tank, wired to the Geneina controller.
FER-008	Manual tank-selection manifold	1 set	Three independent suction/return paths, six full-bore valves, six unions and three return check valves.
FER-009	External loop piping	45 m	1 in uPVC/PVC-U Schedule 80 including compatible fittings, supports, labels and flushing points.
FER-010	Handheld EC/pH/temperature meter	1	Independent daily verification meter with replaceable probes.
FER-011	Calibration and care kit	1 set	pH 4.00, pH 7.00, EC standard, storage/cleaning solutions, cups and calibration log.
FER-012	Preparation table	1	1.50 x 0.75 x 0.90 m high; SS304 or chemical-resistant industrial-plastic top.
FER-013	Digital fertilizer scale	1	Minimum 30 kg capacity and 1 g readability, with calibration weight/check.
FER-014	Service sink and treated-water point	1 set	Chemical-resistant sink, 1/2 in treated-water tap and independent trapped drain.
FER-015	Small protective canopy	1 set	Open-sided weather canopy over skid, stock tanks and local panel; service access maintained.
FER-016	Operational safety and labeling set	1 lot	Removable acid drip tray, eyewash bottle, spill kit, PPE, line tags, arrows and 6 kg dry-powder extinguisher.

3. Hydraulic Arrangement

Path	Diameter/material	Flow sequence
Common side-stream suction	1 in Schedule 80	Selected tank -> branch isolation -> union -> common suction -> internal strainer/pump/flow cell.
Common return	1 in Schedule 80	Geneina skid discharge -> selected return branch -> check valve -> tank return below liquid level.
East production branch	1.5 in tank nozzle / 1 in loop	Dedicated suction and return; no common liquid path to the west tank.
West production branch	1.5 in tank nozzle / 1 in loop	Dedicated suction and return; no common liquid path to the east tank.
Nursery branch	1 in	Independent suction and return to the 3 m3 nursery tank.
Stock A/B/acid suction	Supplier chemical tube	Each stock tank has a dedicated foot valve, filter and dosing channel.
Preparation water	1 in header / 3/4 in top-fill hose	Treated water fills calibrated A/B tanks; acid tank is filled only by controlled chemical transfer.

- External supporting pipework totals 45 m of 1 in uPVC/PVC-U Schedule 80.
- Provide four sampling valves: three served tanks and one after the flow cell.
- Each return branch has a non-return valve; all unions remain accessible.
- Use PE/uPVC for A/B contact and PE/PVDF/PTFE for direct acid contact.

4. Controller Recipes and Operating Limits

Recipe	EC target	EC band	pH target	pH band	Temperature
Production East	1.40 mS/cm	1.20-1.80 mS/cm	5.80	5.50-6.20	19-21 degC
Production West	1.40 mS/cm	1.20-1.80 mS/cm	5.80	5.50-6.20	19-21 degC
Nursery NFT	1.00 mS/cm	0.80-1.20 mS/cm	5.80	5.50-6.20	19-21 degC

Initial commissioning values are refined by the project agronomist. Recommended alarms: EC deviation over 0.30 mS/cm for ten minutes, pH below 5.20 or above 6.50, and temperature above 23 degC.

Condition	Controller response
Tank selection mismatch	Do not enable Auto dosing until the manual manifold line-up and controller recipe are jointly confirmed on the operator checklist.
No side-stream flow	Stop all dosing pumps and alarm; dosing requires proven recirculation flow.
Stock A/B/acid low level	Stop the related dosing pump and alarm without affecting unrelated stock channels.
EC below target	Pulse A, mix and read; then pulse B, mix and read. A and B never inject simultaneously.
EC at target	Inhibit A/B dosing and continue mixing until reading is stable.
pH above target	Pulse acid only after EC has stabilized; apply adjustable mixing delay after every pulse.
pH below low limit	Stop acid dosing, alarm and require operator acknowledgement.
High EC or pH out of safe band	Stop dosing, continue recirculation and require controlled dilution/correction procedure.
Sensor disagreement	Hold automatic dosing when fixed and handheld readings differ by more than accepted tolerance.
Emergency stop	Stop side-stream and dosing pumps; retain alarm/status indication.

5. Stock-Solution Preparation

- Verify the approved fertilizer recipe, batch volume, chemical identity, expiry and PPE before opening any container.
- Confirm Stock A and Stock B tanks are clean, empty and correctly labeled. Never prepare A and B in the same tank or utensil.
- Fill the selected stock tank to approximately 60-70% of the calibrated batch volume with treated water.
- Start the correct A or B mixer. Weigh each fertilizer separately using the 30 kg / 1 g scale and add gradually below the liquid surface.
- Mix until completely dissolved, then add treated water to the calibrated final volume. Record actual mass, water volume, date and operator.
- For acid, use controlled transfer into the dedicated 100 L tank. No water shall be poured into concentrated acid; follow the chemical supplier SDS.
- Close lids, clean external surfaces, place dedicated utensils in their labeled location and update the Fertilizer Preparation Log.

Chemical segregation: use dedicated labeled utensils for A, B and acid. Calcium-containing fertilizer is normally segregated from sulfate/phosphate concentrates. Never pour water into concentrated acid.

6. Tank Correction and Dosing Sequence

- Select one operating tank only at the manual suction/return manifold; close and positively verify the other two circuits.
- Select the matching controller recipe and record the initial tank volume, EC, pH and temperature.
- Start the selected tank NFT pump and Geneina side-stream pump; prove stable flow for at least 60 seconds.
- Dose Stock A in short adjustable pulses, allow 60-120 seconds mixing, then dose Stock B in separate pulses with the same stabilization delay.
- When EC is stable at target, enable acid pulses. Apply the configured delay after each pulse and prevent acid overshoot.
- Continue recirculation for at least five minutes after the final dose, then verify EC/pH/temperature with the handheld meter.
- Record final values and valve line-up before returning the system to normal NFT operation or selecting another tank.

Minimum internal package duties: A and B channels 0-50 L/h each; acid 0-5 L/h; repeatability within +/-5% after calibration. Side-stream flow is 3-5 m³/h and shall be proven before dosing.

7. Installation Sequence

- Release the protected service area, treated-water point, independent drain, power and earth.
- Receive the Geneina package against the signed inclusion list before procuring supporting items.
- Install stock tanks, mixers, table, sink, removable acid tray and canopy with clear service access.
- Install the three-circuit manifold and 1 in Schedule 80 loops; label tanks, valves and flow directions.
- Install suction sets, low-level switches and sampling valves; flush before chemical connection.
- Complete power/control and calibrate sensors, dosing channels and level switches.

8. Testing, Commissioning and Acceptance

Test	Method	Acceptance
Identity/material	Verify package list, tanks, mixers, pipe rating and chemical compatibility.	Accepted material release; no duplicated internal package items.
External loop pressure	Hydrotest clean-water piping at 1.5 times operating pressure or lower component limit.	No leakage or unexplained pressure loss.
Cross-connection proof	Water-test each selected tank path while the other two remain isolated.	Flow returns only to the selected tank.
Dosing calibration	Timed drawdown for A, B and acid channels at representative settings.	Measured dose within +/-5% of controller command.
Sensor calibration	Two-point pH, EC standard and temperature comparison.	Within supplier tolerance and handheld verification.
Low-level protection	Simulate low stock level on A, B and acid tanks.	Only the related dosing channel stops and alarms.
Recipe/selection	Run all three recipes using clean water or approved commissioning solution.	Correct line-up, targets, delays and no unintended dosing.
Integrated correction	Correct one production tank and the nursery tank from known starting values.	Stable target after final five-minute mixing and independent verification.

8.1 Inspection and Test Plan

No.	Inspection stage	Method	Type	Record
01	Supplier package and signed inclusion list	Document review	H	Package release
02	Canopy, pad, sink/drain, treated-water and power interfaces	Survey/visual	H	Work-front release
03	Stock tanks, mixers, manifold, pipe and chemical materials	Identity/certificates	W	Material release
04	External loop pressure and cross-connection tests	Hydrotest/flow test	H	Hydraulic test pack
05	Dosing pump and sensor calibration	Instrumented test	H	Calibration sheets
06	Recipe, alarm and interlock cause-and-effect	Functional test	H	Control acceptance
07	Integrated EC/pH correction and handheld comparison	Performance test	H	Commissioning report
08	Training, logs, spares, O&M and as-builts	Record review	R	Turnover dossier

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

8.2 Hold points

- HP-FER-01: Signed package inclusion list and P&ID accepted before procurement.
- HP-FER-02: Service interfaces accepted before installation.
- HP-FER-03: Pressure and cross-connection tests accepted before chemical connection.
- HP-FER-04: Sensor and dosing calibration accepted before nutrient introduction.
- HP-FER-05: All recipes, alarms and line-up procedure accepted before handover.

9. HSE, Records and Handover

Hazard	Mandatory control	Risk
Acid transfer	SDS, eye/face PPE, gloves, apron, drip tray and spill kit	Medium
Dry fertilizer	Dust control, dedicated scoop and gradual below-surface addition	Medium
Mixer	Guard, secure opening, local isolation and no energized hand entry	Medium
Wet electrical	IP-rated equipment, earthing and LOTO	Medium
Wrong tank	Valve checklist, coding, recipe match and one-tank rule	Low
Chemical waste	Independent drain; never to NFT return or soil	Low

Handover includes the signed inclusion list, P&ID, valve schedule, recipes, stock-tank marks, pressure/cross-connection tests, calibration sheets, Fertilizer Preparation Log, valve checklist, fixed SOP, as-built schematic, O&M manuals, warranties, training and critical spares.